

Case Study Report

Design and Development of a Patient-Specific Cranial Plate Using Medical Imaging and Additive Manufacturing Techniques

Case 02 — Fronto-Temporal Lattice Plate (Lateral View)

1. Introduction

1.1 Cranial Defect

Cranial defects are discontinuities in the skull that can occur due to trauma, decompressive craniectomy, tumor resection, congenital abnormalities, or infections. These defects compromise brain protection and can affect a patient's aesthetics and quality of life. Restoration of the cranial vault is achieved through a surgical procedure known as cranioplasty, using implant materials such as PMMA, titanium, or PEEK.

1.2 Need for Patient-Specific Implants (PSI)

Conventionally shaped implants often result in poor anatomical fit and longer surgical time. Patient-specific implants, designed directly from CT imaging data, accurately match the patient's own anatomy — leading to better fit and improved outcomes. This case study demonstrates a patient-specific implant design workflow using open-source software.

2. Objective

- To design a patient-specific cranial implant model from a segmented skull STL.
- To reconstruct a fronto-temporal defect where the mirror-anatomy technique was not applicable, using a primitive-fitting approach instead.
- To generate a lightweight, print-ready implant geometry using hexagonal lattice optimization.
- To demonstrate a complete, low-cost, open-source design workflow suitable for educational and research use.

3. Case Description

A small cranial defect was identified near the top of the skull model, toward the fronto-temporal region. Unlike Case 01, the defect location and surrounding geometry in this case did not allow for a clean mirror-based reconstruction — the contralateral region

either lacked sufficient symmetry or overlapped with adjacent anatomical features, making the standard mirror technique unreliable here.

Instead, a **Fit Primitive** approach was used: a basic square/box primitive was placed over the defect area as a starting base shape. This primitive was then reshaped using Meshmixer's Sculpt brushes — pulling, pushing, and smoothing the surface until it matched the curvature and boundary of the defect site. Once the base shape was aligned and refined, it was converted into a hexagonal lattice pattern to reduce the plate's overall weight. The completed lattice plate was fitted back onto the skull to verify alignment and coverage before finalizing the design.

4. Materials and Software Used

Category	Details
Imaging Data	Segmented skull STL model
Segmentation Software	3D Slicer (open-source)
Design Software	Autodesk Meshmixer (free)
Output Format	STL (Binary)
Intended Fabrication	FDM 3D Printing

5. Design Methodology — Fit Primitive & Sculpt Workflow

- Step 1 — Import STL File** → Import → load the segmented skull STL into Meshmixer.
- Step 2 — Analysis → Inspector (Repair Skull)** Open the Inspector tool from the Analysis panel to automatically detect and repair holes, non-manifold edges, and flipped normals, resulting in a clean, watertight mesh.
- Step 3 — Identify Defect Region** Visually inspect the defect near the fronto-temporal region. Since the surrounding anatomy did not present a reliable symmetric counterpart, the mirror technique was ruled out for this defect.
- Step 4 — Add a Primitive Shape** Use Meshmixer's **Fit Primitive** tool (Edit → Fit Primitive, or Add Primitive) to place a basic square/box shape roughly over the defect area, sized approximately to the defect boundary.
- Step 5 — Position the Primitive** Use the Transform tool to move, rotate, and scale the primitive so it roughly covers the defect region on the skull surface.
- Step 6 — Sculpt the Primitive to Match Skull Curvature** Switch to Sculpt → Brushes (Drag, Smooth, ShrinkWrap) and reshape the flat primitive surface to follow the natural

curvature of the surrounding skull, pulling and pushing the mesh until it wraps closely over the defect area.

Step 7 — Refine Edges Against the Defect Boundary Use the Sculpt → Refine brush along the edges of the shaped primitive to align it precisely with the defect margin, closing any visible gaps against the surrounding bone.

Step 8 — Verify Fit on the Skull Rotate the model and visually inspect the sculpted patch from multiple angles to confirm it sits flush against the defect boundary with no major gaps or overlaps.

Step 9 — Trim Excess Geometry Use Edit → Plane Cut or Trim to remove any excess material extending beyond the intended defect boundary.

Step 10 — Create Hexagonal Lattice Pattern Apply Edit → Make Pattern (Voronoi/Lattice, hexagonal style) to the shaped plate body, reducing weight while retaining structural coverage across the defect.

Step 11 — Add Flanges Use the Draw brush to add small flanges along the plate periphery for fixation and added strength.

Step 12 — Add Cylinders for Screw Holes Position cylinder primitives on the flanges, sized to match standard fixation screws, and apply Boolean Difference to create clean through-holes.

Step 13 — Export STL Run Analysis → Inspector one final time to confirm mesh integrity, then export the finished plate as a binary STL file for printing/further use.

6. Results

The completed design is a patient-specific, hexagonal lattice-structured cranial plate covering the fronto-temporal defect region, built using a primitive-and-sculpt approach rather than mirror reconstruction. The plate was visually verified for alignment and coverage against the original skull model, viewed from the lateral perspective, prior to final export.

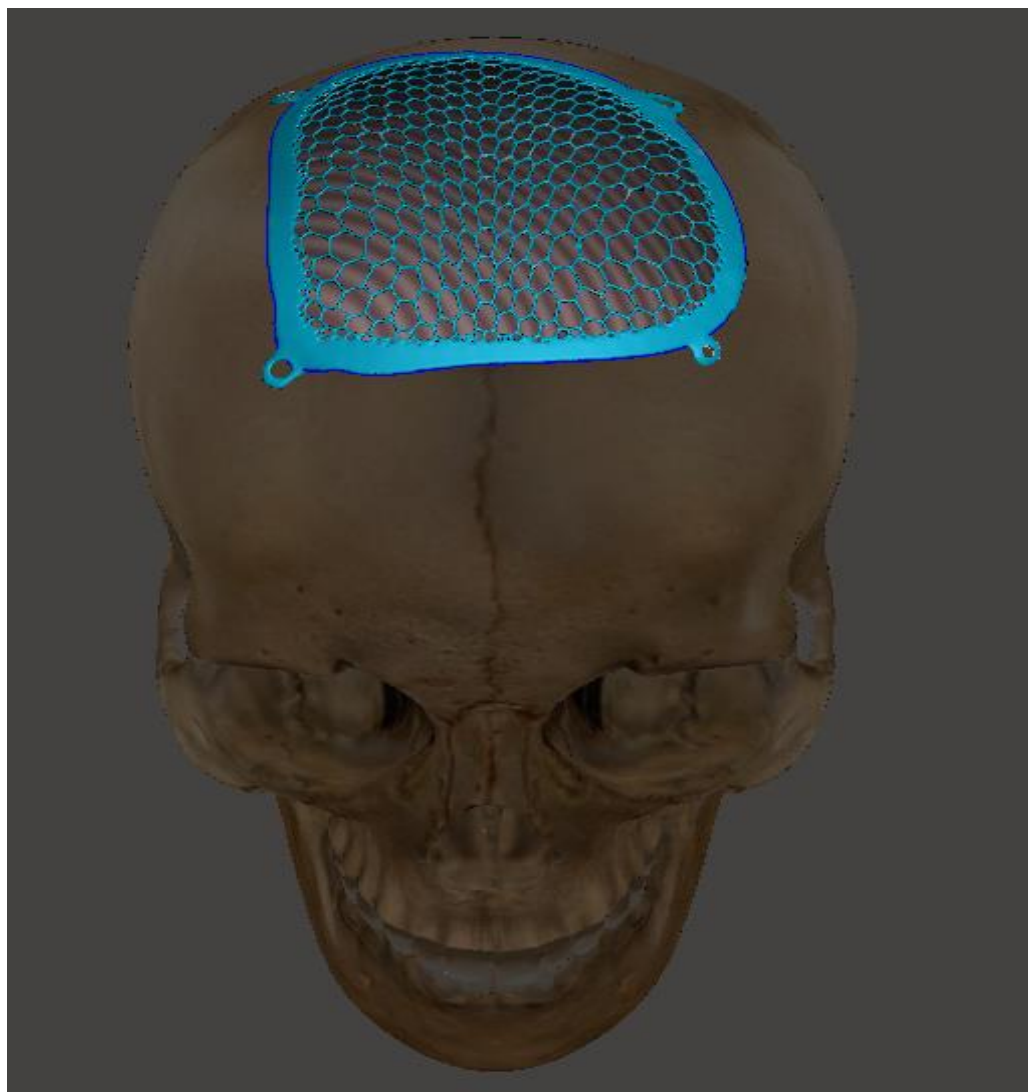
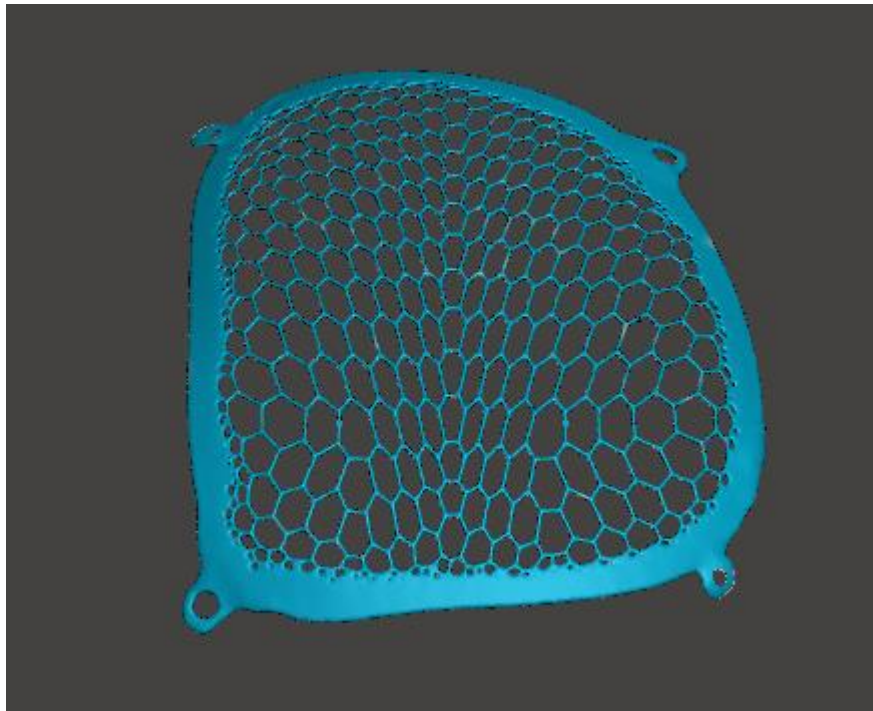
Parameter	Value
Defect Region	Fronto-Temporal
Design Technique	Fit Primitive + Sculpt shaping + Hexagonal lattice
Offset	3–5 mm
Output File	STL (Binary)

Software Used	Meshmixer, 3D Slicer
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7. Conclusion

This case demonstrates that the mirror-anatomy technique, while effective in many cases, is not always applicable — particularly when the defect region lacks a clean symmetric counterpart. Here, a **Fit Primitive combined with manual Sculpt-brush shaping** provided an effective alternative, allowing the plate geometry to be built up manually and refined to match the defect boundary. This highlights the importance of having multiple reconstruction techniques available depending on defect location and anatomy.





Disclaimer: This case study is created purely for **educational and research purposes** as part of the Cranial3DLab project. It is intended to demonstrate the technical design workflow using open-source tools and does not represent a validated clinical implant. Any real-world surgical implant requires professional medical oversight, regulatory approval, and certified manufacturing processes.